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Non-C_2 symmetric chiral ligand dictating allyl-allyl coupling

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Enantioselective synthesis of chiral compounds is of high interest due to their importance in organic chemistry, medicinal chemistry, life science, and materials science. Catalytic construction of chiral centers via the enantioselective carbon−carbon coupling is one of the most efficient approaches. Chiral 1,5-alkadienes are prevalent in natural and bioactive compounds. Thus, allyl-allyl coupling towards to the synthesis of chiral 1,5-alkadienes is of high interest. However, in addition to the issue of enantioselectivity, the extra challenge is the issue of regio-selectivity referring to the branched and linear products caused by the steric and electronic effect of 1,3-terminals. Herein, such challenge is addressed by a synergistic copper/palladium catalysis strategy: A three-component allylboration reaction of allenes with bis(pinacolato)diboron and allylic phosphates is developed to afford a series of optically active 1,5-alkadienes with either tertiary or quaternary carbon stereocenters. Their synthetic potentials are demonstrated, including the synthesis of (-)-protrifenbute and its derivatives.

Transition metal-catalyzed coupling reactions of organic electrophiles with organometallic reagents has been well established to meet the demand for the construction of chiral centers in organic synthesis (Fig. 1a)[1–3]. On the other hand, the allyl-allyl cross-coupling reaction is of great importance since it affords synthetically versatile 1,5-diene products[4]. However, there is an issue of regioselectivity forming the linear and the branched products (Fig. 1b). Due to the steric effect, usually linear products are formed as observed by Trost[5], Stille[6], and their coworkers. Interestingly, the branched 1,5-alkadienes are very common in natural products and the two C=C bonds would also make such products a class of platform molecules for efficient syntheses of natural products and biologically active molecules, for example, (-)-preclamol[7], (-)-protrifenbute[8], (-)-paroxetine[9], rottnestol[10], herboxidiene[11], and α-cuparenone[12] (Fig. 1e)[13–15]. In 2010, the first asymmetric enantioselective coupling between allylic *t*-butyl carbonates and allylBpin, forming such branched products with a tertiary chiral center, was reported by Morken et al. with ee of 74–91%. The regioselectivity for 3-aryl substituted allylic *t*-butyl carbonates is >20:1 while that of 3-alkyl ones is ~10:1[16]. Feringa et al. observed that in the Cu-catalyzed reaction of 2-alkenyl bromide with allyl Grignard reagent[17] in 58–94% ee and 37:63–91:9 branched over linear selectivity as is observed also by Ohmiya/Sawamura in the Cu-catalyzed reaction of 2-alkenylic phosphonates with allylboronates[18] and by Wang/Ding in the Pd-catalyzed reaction of 1-arylallylic acetates with allylboronates[19]. Only in the Ir-catalysis, a high selectivity for both enantio- and branched selectivity has been observed by Carreira and Yang independently in the Ir-catalyzed coupling of allylic alcohols with allyl silanes[20] or boronates[21]. The allyl-allyl coupling for the enantioselective establishment of quaternary carbon stereocenter is even more challenging. The only report is also from Morken and his coworkers: the reaction of racemic tertiary 1,1-disubstituted allylic carbonates with allylboronates afforded the products with a quaternary chiral center in 52–92% ee, and due to the greatly increased steric hinderance, the coupling reactivity is lower, thus, resulting in β-elimination as the side reaction[22]. Since the boracurpation of allenes would provide allylic metallic reagents[23–32], it would be interesting to develop the three-component reaction of differently substituted allenes, B_2Pin_2, and allylic electrophiles

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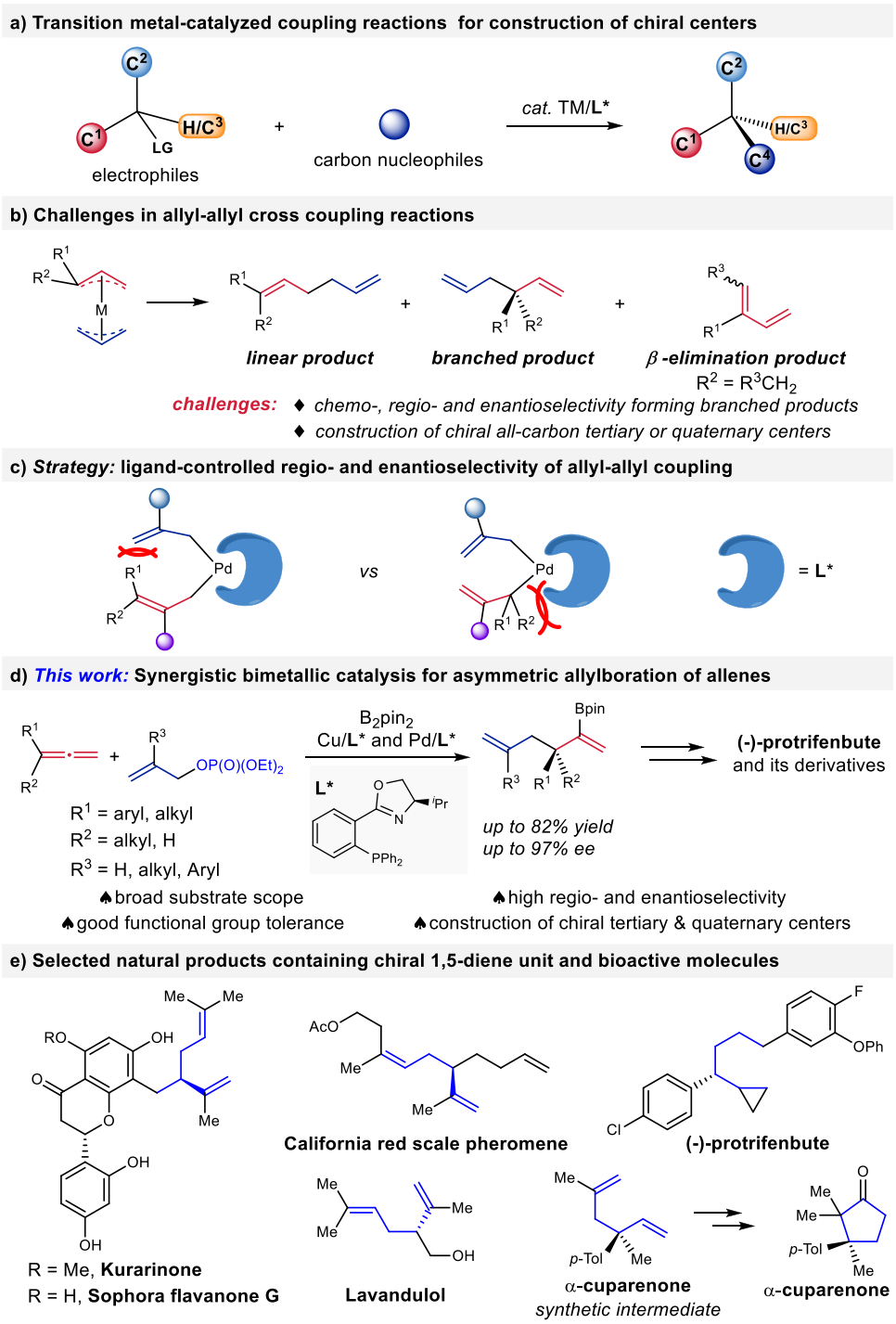


Fig. 1 | Catalytic asymmetric C-C and allyl-allyl couplings as well as selected examples of related natural products. a Transition metal-catalyzed coupling reactions for the construction of chiral centers. **b** Challenges in allyl-allyl cross-coupling reactions. **c** Strategy: ligand-controlled regio- and enantioselectivity of allyl-allyl coupling. **d** This work: Synergistic bimetallic catalysis for asymmetric allylboration of allenes. **e** Selected natural products containing chiral 1,5-diene unit and bioactive molecules.

for such a purpose with an attractive diversity for organic synthesis. Hoveyda and coworkers reported such a three-component reaction of OTBS-substituted 1,2-pentadiene and OTBDPS-substituted 1,2-hexadiene with 3-substituted allylic phosphonates, and the reaction demonstrated a linear selectivity referring to allenes[33], as is also observed by Xiong et al[34]. Fañanás-Mastral et al. observed the first branched selectivity referring to the reaction between mono-substituted allenes and allyl *t*-butyl carbonate, albeit with ~40% ee[35]. One solution could be to identify a finely tuned chiral catalyst with a much higher reactivity, avoiding β-elimination while dictating the branched vs linear selectivity (Fig. 1c).

Here, we wish to disclose our recent observation on the synergistic Cu/Pd bimetallic catalyzed three-component highly regio- and enantioselective allylboration of monoaryl- as well as monoalkyl-substituted allenes with B_2Pin_2 and allylic phosphonates generating chiral 1,5-dien-2-yl boronates with tertiary chiral centers in excellent branched selectivity towards to allenes with an enantioselectivity of

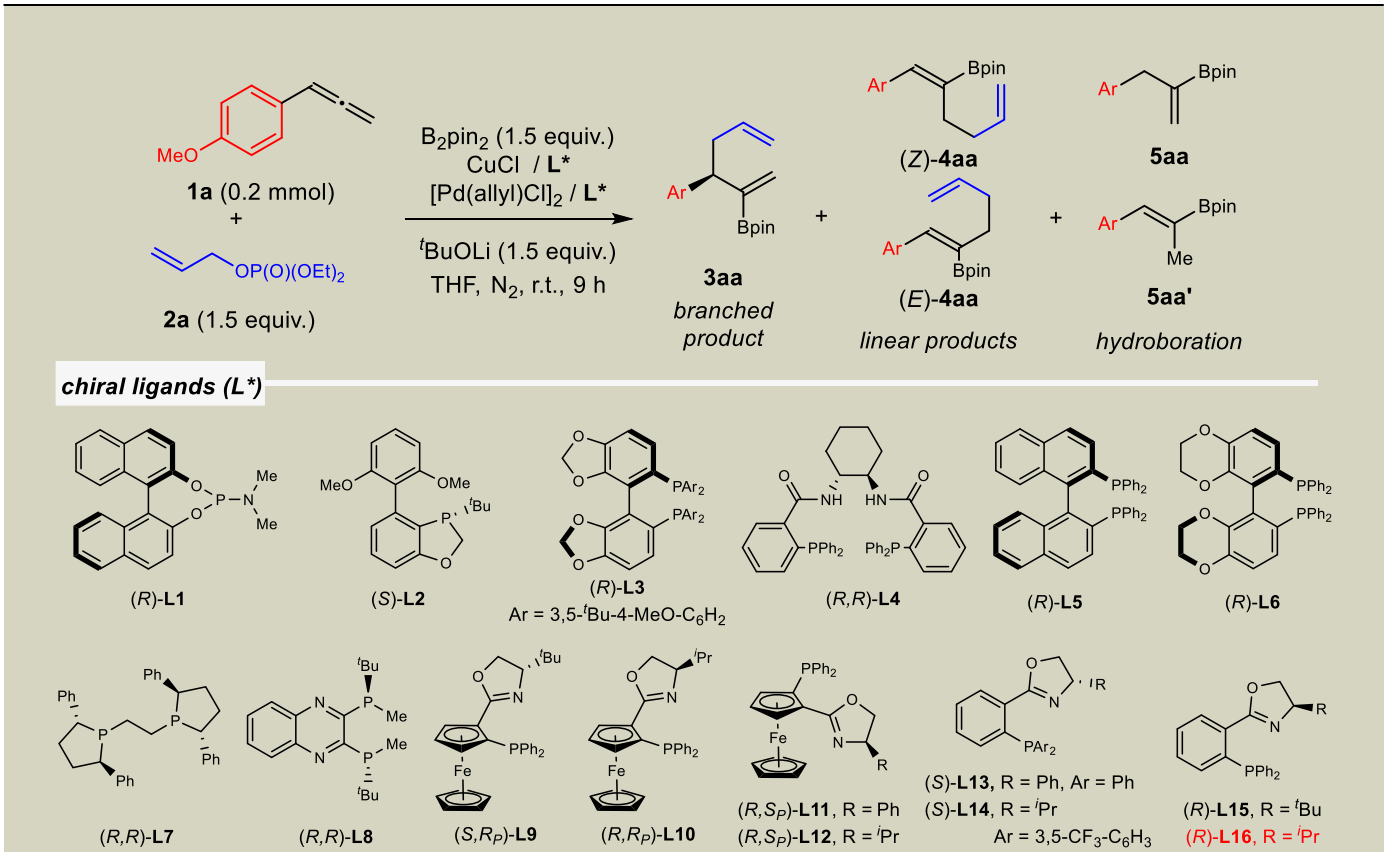
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up to 97% ee (Fig. 1d). (-)-Protrifenbute and its two derivatives are synthesized efficiently by using this protocol as the key step. Furthermore, it is exciting to observe that such a strategy could also be applied to the construction of much more challenging all-carbon quaternary stereocenters as well when applying 1,1-disubstituted allenes. As is known, the construction of all-carbon quaternary stereocenters is still a challenge due to the requirement of steric bias of the four substituents.

Results

To avoid the issue of β-elimination, at the beginning, *p*-methoxyphenylpropadiene **1a**, B_2pin_2, and allyl diethyl phosphate **2a** were chosen as the model substrates to develop the asymmetric three-component reaction with C_2-symmetric chiral ligand (Table 1). Firstly, we used CuCl and $[Pd(allyl)Cl]_2$ as metallic catalyst precursors[36–47] and ^{t}BuOLi as the base to screen typical chiral ligand skeletons in THF at room temperature: When mono-dentated phosphoramidite ligand (*R*)-

Table 1 | Selected representative results of ligand screening



entry	L*	Recovery of 1a (%)	Yield/ee[a] of 3aa (%)	Yield of 4aa (%)/(Z:E)	Yield of 5aa/5aa′ (%)
1	(*R*)-**L1**	72	ND/-	21/(67:33)	12/8
2[b]	(*S*)-**L2**	53	ND/-	8/(75:25)	3/5
3	(*R*)-**L3**	93	ND/-	ND/-	ND/ND
4	(*R,R*)-**L4**	96	ND/-	ND/-	ND/ND
5	(*R*)-**L5**	64	5/-	trace/-	7/ND
6	(*R*)-**L6**	41	33/−24	4/(0:100)	13/13
7	(*R,R*)-**L7**	24	54/48	17/(76:24)	trace/trace
8	(*R,R*)-**L8**	27	28/−84	8/(100:0)	10/trace
9	(S,R_P)-**L9**	53	18/−56	23/(83:17)	ND/ND
10	(R, R_P)-**L10**	ND	97/72	4/(100:0)	2/ND
11	(R,S_P)-**L11**	14	61/59	22/(86:14)	3/ND
12	(R, S_P)-**L12**	31	59/81	5/(40:60)	ND/ND
13	(*S*)-**L13**	38	20/−54	27/(70:30)	2/3
14	(*S*)-**L14**	36	13/-50	33/(76:24)	ND/ND
15	(*R*)-**L15**	8	34/66	57/(72:28)	3/2
16	(*R*)-**L16**	ND	83/89	18/(50:50)	ND/ND

Reaction conditions: **1a** (0.2 mmol, 1.0 equiv.), **2a** (1.5 equiv.), B_2pin_2 (1.5 equiv.), ^{t}BuOLi (1.5 equiv.) CuCl (5 mol%)/**L*** (6 mol%), $[Pd(allyl)Cl]_2$ (2.5 mol%)/**L*** (6 mol%) THF (2.0 mL, 0.1 M), N_2, r.t., 9 h. Yields and the recoveries were determined by ^{1}H NMR analysis using CH_3NO_2 as the internal standard.

ND not detected.

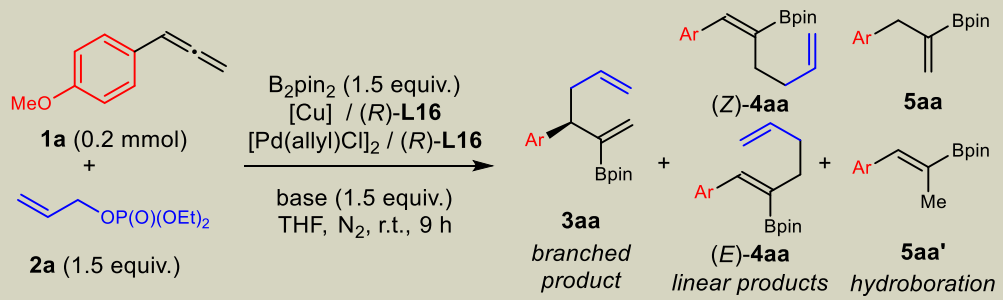
[a]Determined by HPLC after oxidation of **3aa** with $NaBO_3{\cdot}4H_2O$.

[b]**L*** (12 mol%).

L1[48], monophosphine ligand (*S*)-**L2**, biphenyl-based bisphosphine ligands (*R*)-**L3**, or Trost ligand (*R,R*)-**L4** was applied, no branched product **3aa** was generated (Table 1, entries 1–4); the linear coupling products (*Z*)-**4aa** and (*E*)-**4aa** as well as hydroboration products **5aa** and **5aa'** were detected in presence of (*R*)-**L1** or (*S*)-**L2** as the ligand as expected (Table 1, entries 1 and 2). Interestingly, with BINAP (*R*)-**L5** as the ligand, the targeted branched coupling product **3aa** was afforded albeit in 5% NMR yield (Table 1, entry 5); with SYNPHOS (*R*)-**L6** and BPE-PHOS (*R,R*)-**L7**, the desired product **3aa** was obtained in 33% yield and -24% ee, 54% yield and 48% ee, respectively (Table 1, entries 6 and 7); interestingly, Quinox P* (*R,R*)-**L8** improved the ee to 84%, albeit still merely in 28% yield (Table 1, entry 8). To our delight, when *non*-C_2-symmetric phosphine-oxazoline chiral ligands **L9**-**L16** were applied (Table 1, entries 9–16), (*R*)-**L16** was identified as the optimal chiral ligand affording the product **3aa** smoothly with 83% NMR yield and 89% ee (Table 1, entry 16).

However, the branched/linear selectivity is still a problem (Table 1, entries 6-16). Thus, (*R*)-**L16** was chosen as the ligand to optimize other reaction parameters (Table 2). Firstly, using $[Pd(allyl)Cl]_2$ as the palladium catalyst precursor, several copper salts were screened (Table 2, entries 1–5) and CuTc was identified to be the best (Table 2, entry 4). Then with CuTc, $PdCl_2$, $Pd_2(dba)_3$, and $Pd(OAc)_2$ were tested, however,

Table 2 | Optimization of reaction conditions



Entry	[Cu]	base	Recovery of 1a (%)	Yield/ee[a] of 3aa (%)	Yield of 4aa (%)/(Z:E)	Yield of 5aa/5aa' (%)
1	CuCl	^{t}BuOLi	ND	83/89	18/(50:50)	ND/ND
2	CuOAc	^{t}BuOLi	ND	72/85	26/(65:35)	3/3
3	$Cu(OAc)_2$	^{t}BuOLi	ND	75/85	25/(64:36)	trace/trace
4	CuTc	^{t}BuOLi	ND	85/88	22/(55:45)	ND/ND
5	CuBr	^{t}BuOLi	ND	78/87	21/(57:43)	trace/trace
6[b]	CuTc	^{t}BuOLi	ND	40/59	45 / (87:13)	8/6
7[c]	CuTc	^{t}BuOLi	ND	73/88	19/(53:47)	8/4
8[d]	CuTc	^{t}BuOLi	ND	56/79	29/(69:31)	11/7
9	CuTc	^{t}BuOLi	ND	13/47	8/(50:50)	5/13
10	CuTc	^{t}BuONa	ND	46/85	12/(42:58)	trace/trace
11	CuTc	CsF	4	70/90	14/(50:50)	14/11
12	CuTc	MeOLi	ND	74/92	12/(58:42)	9/13
13[e]	CuTc	MeOLi	34	42/91	8/(38:62)	9/12
14[f]	CuTc	MeOLi	43	40/93	4/(50:50)	3/4
15[g]	CuTc	MeOLi	ND	73/92	11/(45:55)	8/13
16[h]	CuTc	MeOLi	ND	70/90	12/(67:33)	11/14
17[i]	CuTc	MeOLi	ND	76/95	6/(67:33)	6/13
18[j]	CuTc	MeOLi	4	77/96	3/(67:33)	5/9
19[j,k]	CuTc	MeOLi	26	47/96	3/(67:33)	5/15
20[f,l]	CuTc	MeOLi	2	74/94	4/(50:50)	3/5
21[m]	CuTc	MeOLi	ND	87 (70)[n]/96	5/(60:40)	3/7

Reaction conditions: **1a** (0.2 mmol, 1.0 equiv.), **2a** (1.5 equiv.), B_2pin_2 (1.5 equiv.), base (1.5 equiv.) [Cu] (5 mol%)/(*R*)-**L16** (6 mol%), $[Pd(allyl)Cl]_2$ (2.5 mol%)/(*R*)-**L16** (6 mol%), solvent (2.0 mL, 0.1 M), N_2, r.t., 9 h. Yields and the recoveries were determined by ^{1}H NMR analysis using CH_3NO_2 as the internal standard.
ND not detected.
[a]Determined by HPLC after oxidation of **3aa** using $NaBO_3{\cdot}4H_2O$.
[b]$PdCl_2$ (5.0 mol%) instead of $[Pd(allyl)Cl]_2$ (2.5 mol%).
[c]$Pd_2(dba)_3$ (2.5 mol%) instead of $[Pd(allyl)Cl]_2$ (2.5 mol%).
[d]$Pd(OAc)_2$ (5.0 mol%) instead of $[Pd(allyl)Cl]_2$ (2.5 mol%).
[e]DCM instead of THF.
[f]DCE instead of THF.
[g]1,4-dioxane instead of THF.
[h]Toluene instead of THF.
[i]0.2 M, 0 °C, 18 h.
[j]**1a** (1.0 mmol scale, 0.5 M), −10 °C, 24 h.
[k]CuTc (2.5 mol%)/(*R*)-**L16** (3 mol%), $[Pd(allyl)Cl]_2$ (1.25 mol%)/(*R*)-**L16** (3 mol%).
[l]**1a** (1.0 mmol scale, 0.5 M), r.t., 38 h.
[m]**1a** (1.0 mmol scale, 0.25 M); **1a**, **2a**, and B_2pin_2 were dried over 5 Å MS before use, respectively; −10 °C, 36 h.
[n]Isolated yield.

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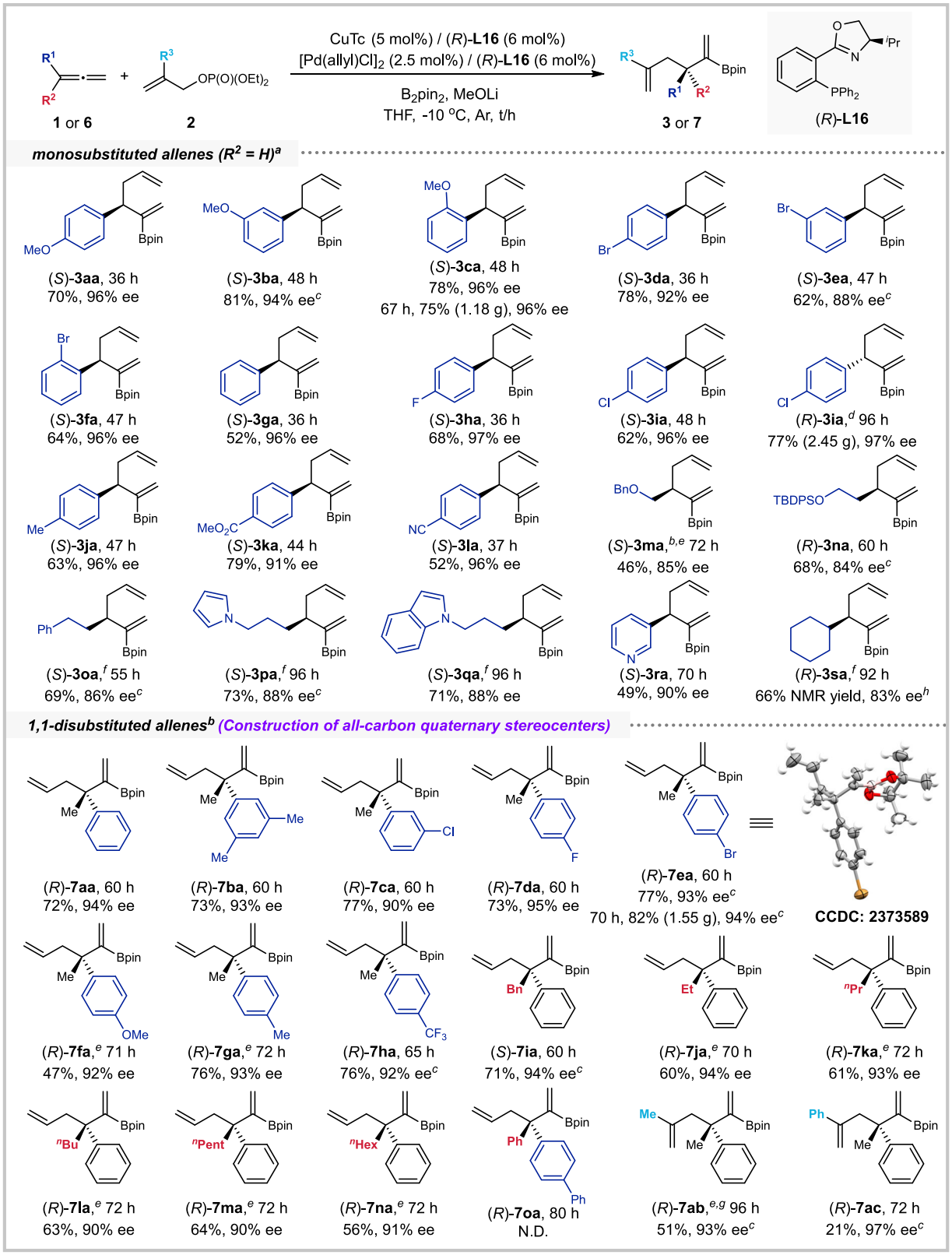


Fig. 2 | Substrate scope. [a] Conditions **A**: **1** (1.0 mmol, 1.0 equiv.), **2** (1.5 equiv.), B_2Pin_2 (1.5 equiv.), MeOLi (1.5 equiv.), CuTc (5 mol%)/(*R*)-**L16** (6 mol%), $[Pd(allyl)Cl]_2$ (2.5 mol%)/(*R*)-**L16** (6 mol%), THF (4.0 mL, 0.25 M), −10 °C. [b] Conditions **B**: **1** or **6** (1.0 mmol, 1.0 equiv.), **2** (2.0 equiv.), B_2Pin_2 (2.0 equiv.), MeOLi (2.0 equiv.), CuTc (5 mol%)/(*R*)-**L16** (6 mol%), $[Pd(allyl)Cl]_2$ (2.5 mol%)/(*R*)-**L16** (6 mol%), THF (4.0 mL, 0.25 M), −10 °C. [c] determined by HPLC analysis after oxidation of **3** or **7** using $NaBO_3 \cdot 4H_2O$. [d] **1i** (10.0 mmol, 1.0 equiv.), **2a** (1.2 equiv.), (*S*)-**L16** was used instead of (*R*)-**L16**. [e] MeOLi (4.0 equiv.). [f] MeOLi (3.0 equiv.). [g] $[Pd(2\text{-methylallyl})Cl]_2$ instead of $[Pd(allyl)Cl]_2$. [h] determined by HPLC analysis after being converted to (*R*)-**10sa**, for details, see Supplementary Information. ND not detected.

no better results were obtained (Table 2, entries 6–8). Since the base may play a critical role, different bases were applied (Table 2, entries 9–12): When MeOLi was used, the target compound **3aa** was obtained with 74% yield and 92% ee. However, more hydroboration products **5aa** and **5aa'** were formed (Table 2, entry 12). Subsequently, with MeOLi as the base, the solvent effect was explored and high yields and selectivity were achieved in tetrahydrofuran (Table 2, entries 12–16). When the reaction was conducted at 0 °C, the chemo-, regio-, and enantioselectivity were improved (Table 2, entry 17). Better results were obtained at −10 °C with the concentration of 0.5 M (Table 2, entry 18). A lower catalyst loading led to 47% NMR yield (Table 2, entry 19). The reaction at a concentration of 0.5 M was too viscous (entry 18), which may have a negative effect on the reaction yield. We reasoned that the trace amount of the water in the ambient environment may result in the formation of the hydroboration products **5aa** and **5aa'**, therefore, all the starting materials **1a, 2a** and B_2pin_2 were dried over 5 Å molecular sieves before use and the branched product **3aa** was obtained with 96% ee, 70% isolated yield, and a remarkable branched/linear selectivity of 95/5 at a concentration of 0.25 M, which has been defined as the optimized conditions (Table 2, entry 21).

The optimized reaction conditions (Table 2, entry 21) have then been applied to evaluate the scope of the allenes and allylic phosphates. This synergistic Cu/Pd enantioselective catalysis protocol tolerates a range of monosubstituted and 1,1-disubstituted allenes, generating a series of boron contained chiral 1,5-dienes **3** or **7** with moderate to good yields and excellent regio- and enantioselectivity (Fig. 2). The substituent location in the phenyl group has no obvious effect since both the *o*- or *m*-$MeOC_6H_4$ substituted allenes **1b** and **1c** gave corresponding products (*S*)-**3ba** and (*S*)-**3ca** with 94 and 96% ee, respectively. Even the *o*-, *m*-, and *p*-bromophenyl substituted allenes **1d, 1e**, and **1f** afforded (*S*)-**3da**, (*S*)-**3ea**, and (*S*)-**3fa** without touching the reactive and versatile C-Br bond. The gram-scale preparation of chiral 1,5-diene (*S*)-**3ca** could be achieved readily in 75% yield and 96% ee. A variety of synthetically useful functional groups, such as F, Cl, Me, CO_2Me, and CN, were accommodated, affording (*S*)-**3ha** - (*S*)-**3la** successfully with ee ranging from 91 to 97%. As expected, when (*S*)-**L16** was introduced instead of (*R*)-**L16**, the corresponding enantiomer (*R*)-**3ia** was produced on a 10 mmol scale with 97% ee and 77% isolated yield with only 1.2 equiv of **2a**. Besides the monoaryl-substituted substrates, monoalkyl-substituted allenes also worked to afford (*S*)-**3ma**, (*R*)-**3na**, and (*S*)-**3oa** with moderate yields and a decent enantioselectivity of 84 to 86% ee. The reaction also tolerated heteroaryl groups, such as pyrrole (**1p**), indole (**1q**), and pyridine (**1r**), affording the corresponding products (*S*)-**3pa**, (*S*)-**3qa**, and (*S*)-**3ra** with 88% to 90% ees in 49 to 73% yields. Employing

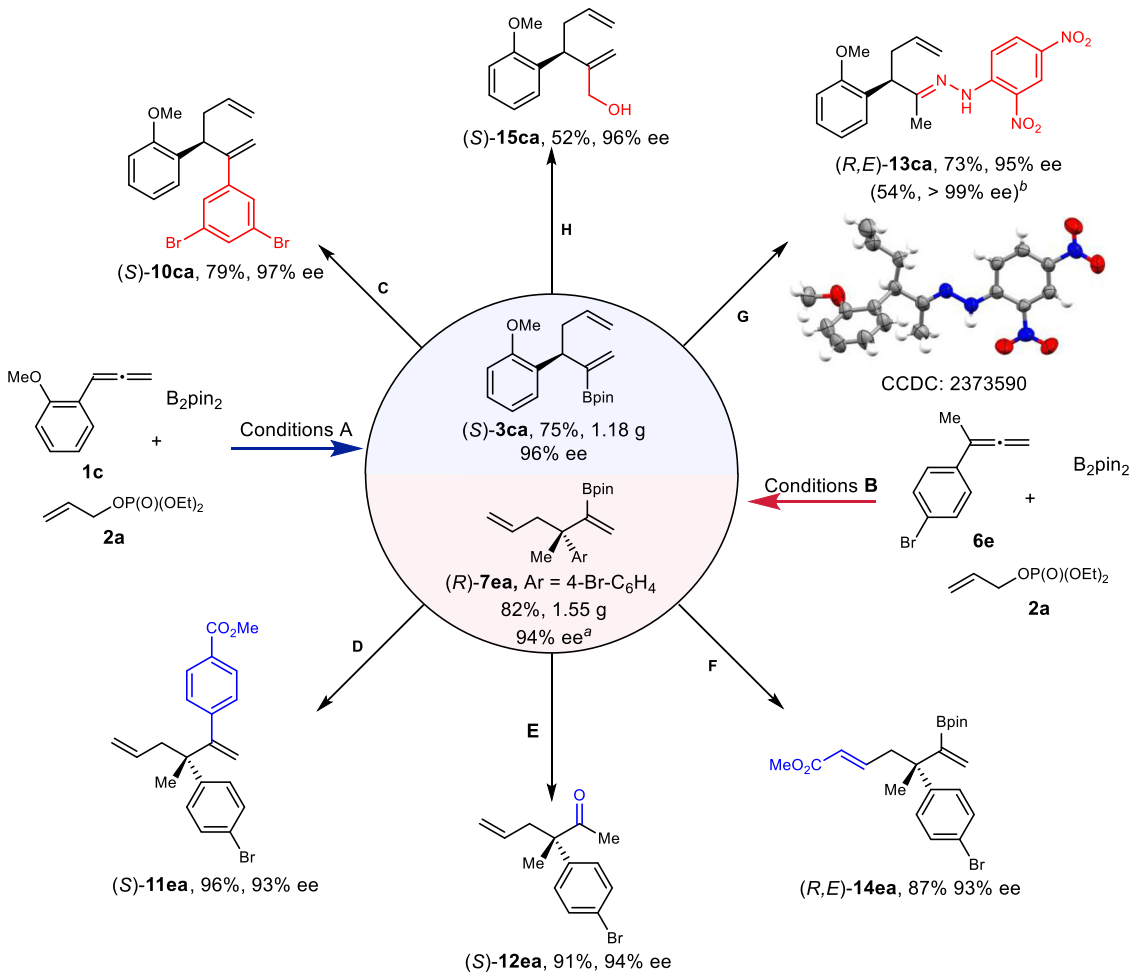


Fig. 3 | Gram-scale synthesis and transformations of (*S*)-3ca and (*R*)-7ea. Conditions **A** and **B**: see Fig. 2. [a] determined by HPLC analysis after oxidation using $NaBO_3$•$4H_2O$. **C**: (*S*)-**3ca** (1.0 equiv.), 1,3-dibromo-5-iodobenzene (3.0 equiv.), $Pd(PPh_3)_4$ (10 mol%), Cs_2CO_3 (3.0 equiv.), THF (2.0 mL), 80 °C, 24 h. **D**: (*R*)-**7ea** (1.0 equiv.), methyl 4-bromobenzoate (3.0 equiv.), $Pd(PPh_3)_4$ (10 mol%), Cs_2CO_3 (3.0 equiv.), THF (2.0 mL), 80 °C, 59 h. **E**: (*R*)-**7ea** (1.0 equiv.), $NaBO_3$•$4H_2O$ (5.0 equiv.), THF (5.0 mL), H_2O (5.0 mL), r.t., 5 h. **F**: (*R*)-**7ea** (1.0 equiv.), methyl acrylate (3.0 equiv.), second generation Grubbs catalyst (5 mol%), CuI (7.5 mol%), Et_2O (2.0 mL), 35 °C, 7 h. **G**: (*S*)-**3ca** (1.0 equiv.), $NaBO_3$•$4H_2O$ (5.0 equiv.), THF (5.0 mL), H_2O (5.0 mL), r.t., 3 h, then (2,4-dinitrophenyl)hydrazine (1.0 equiv.), BF_3•Et_2O (1.0 equiv.), MeOH (2.0 mL), −78 °C-r.t., 6 h. [b] after recrystallization in DCM/PE. **H**: (*S*)-**3ca** (1.0 equiv.), ICH_2Cl (1.5 equiv.), ^{n}BuLi (1.1 equiv.), THF (1.0 mL), −78 °C-r.t., 16.5 h, then, NaOH (2.5 M, aq., 0.5 mL), H_2O_2 (30%, 0.5 mL), 0 °C, 12 h.

cyclohexyl substituted allene **1s** as the starting material, which had the similar steric hindrance to the phenyl substituted allene **1g**, target product (*R*)-**3sa** was obtained in 66% NMR yield with 83% ee, which was similar to those of other alkyl-substituted allenes [(*R*)-**3na** - (*S*)-**3oa** (84-86% ees)]. These results ruled out that the steric hindrance led to the difference for aryl- and alkyl-substituted allenes, and we reasoned that the electronic effect may be the key point.

Furthermore, even all-carbon quaternary stereocenters may be constructed by applying the 1,1-disubstituted allenes forming 1,5-pentadiene **7** with excellent levels of regio- and enantioselectivity: the reaction of 3-phenylbuta-1,2-diene **6a** afforded the desired branched product (*R*)-**7aa** with 72% isolated yield and 94% ee. It worth to mention that the linear products and hydroboration products were not detected, and only a trace amount of β-H elimination product (<4% NMR yield, if any) was detected by ^{1}H NMR analysis of crude products (see Supplementary Information). Synthetically versatile F, Cl, Br, Me, and methoxy group were all tolerated on the phenyl ring of the 1,1-disubstituted allenes affording (*R*)-**7ba** - (*R*)-**7ha** with moderate yields and excellent ee values (Fig. 2). The gram-scale synthesis of (*R*)-**7ea** has been achieved easily (1.55 g, 82% isolated yield and 94% ee). Furthermore, the absolute configuration of the all-carbon quaternary stereocenters was determined by the X-ray single crystal diffraction analysis of (*R*)-**7ea**. Methyl or methoxy-substituent within the phenyl ring also worked, requiring a higher loading of MeOLi to ensure a high yield (Fig. 2, (*R*)-**7fa** and (*R*)-**7ga**). Besides, 3,4-diphenylbuta-1,2-diene **6i** may also afford the target product (*S*)-**7ia** with 71% yield and 94% ee. The methyl group may be replaced with Et, ^{n}Pr, ^{n}Bu, ^{n}Pent, or ^{n}Hex group (**6a, 6j-n**), affording (*R*)-**7ja**, (*R*)-**7ka**, (*R*)-**7la**, (*R*)-**7ma**, or (*R*)-**7na** with an excellent ee. However, when 1,1-diaryl-substituted allene **6o** was employed, the corresponding product (*R*)-**7oa** was not detected. Diethyl 2-methylallyl phosphate **2b** and diethyl 2-phenylallyl phosphate **2c** may also be applied to afford the desired products (*R*)-**7ab** and (*R*)-**7ac** in 93% and 97% ee, respectively.

The synthetic potentials of these chiral 1,5-pentadienes have been demonstrated (Fig. 3): the gram-scale reactions of **1c** and **6e** with allyl diethyl phosphate **2a** under the standard conditions afforded (*S*)-**3ca** and (*R*)-**7ea** smoothly, which readily underwent Suzuki coupling[49] selectively with the C-I bond in 1,3-dibromo-5-iodobenzene and the C-Br bond in methyl 4-bromobenzoate affording the arylation products (*S*)-**10ca** and (*S*)-**11ea**, respectively; oxidation of the alkenyl boronate unit in (*R*)-**7ea** with $NaBO_3$•$4H_2O$ results in the synthesis of α-chiral homoallyl ketone (*S*)-**12ea** with 91% yield and 94% ee; oxidation of (*S*)-**3ca** followed by condensation with (2,4-dinitrophenyl)hydrazine[50] led to the formation of optically active solid product (*R*,*E*)-**13ca** in 73% yield and 95% ee, and its X-ray single crystal diffraction analysis further confirmed the stereoselectivity of this reaction. Catalytic cross metathesis reaction of (*R*)-**7ea** occurred highly selectively with mono-substituted terminal C=C bond, furnishing functionalized chiral 2,5-dienoate (*R*,*E*)-**14ea** with 87% isolated yield and 93% ee[51]; the reaction of (*S*)-**3ca** with ^{n}BuLi followed by the addition of ICH_2Cl and hydrolysis afforded 1,5-dienol (*S*)-**15ca** in 52% yield and 96% ee[52].

To further demonstrate the synthetic potential of this synergistic Cu/Pd catalysis strategy, the synthesis of (-)-protrifenbute[8] has been executed (Fig. 4). Firstly, (*R*)-**3ia** was prepared smoothly by following this protocol (for details, see Fig. 2); upon selective catalytic olefin

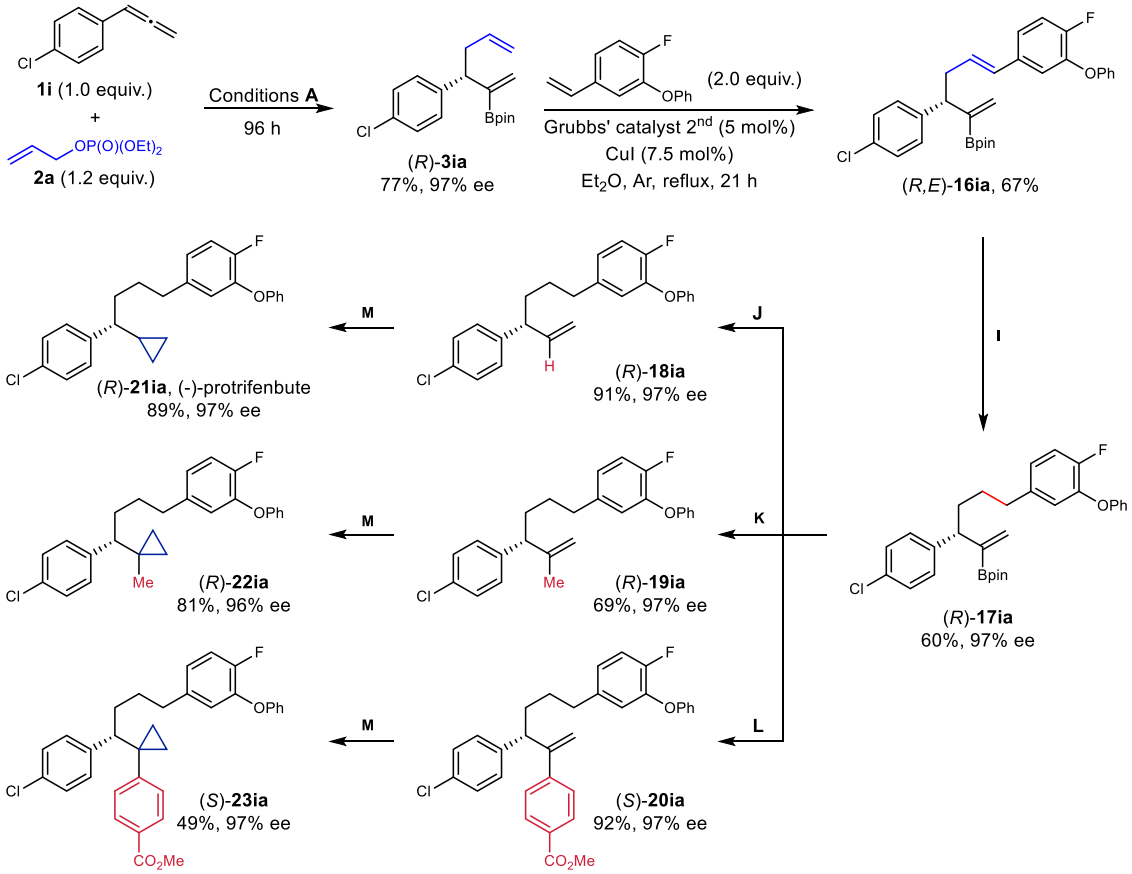
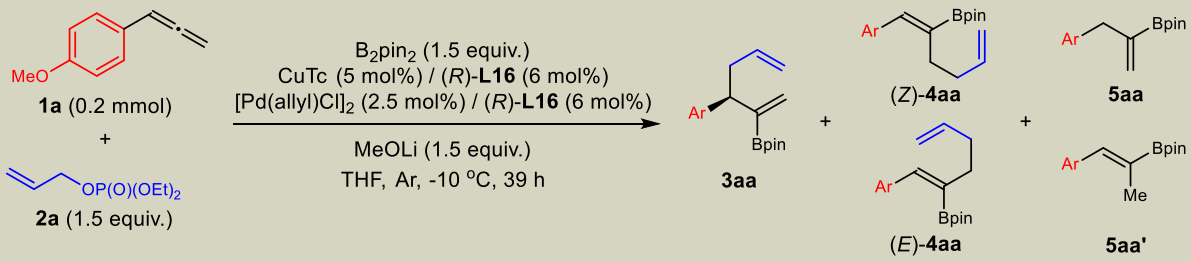


Fig. 4 | Synthesis of (-)-protrifenbute and its derivatives. Conditions **A**: see Fig. 2. **I**: (*R*,*E*)-**16ia** (1.0 equiv.), 5% Pd/C (5 mol%), $(MeO)_2MeSiH$ (3.0 equiv.), EtOH, air, r.t., 4.5 h. **J**: (*R*)-**17ia** (1.0 equiv.), AgF (2.1 equiv.), THF/MeOH/H_2O (10/9/1), Ar, 50 °C, 6 h. **K**: (*R*)-**17ia** (1.0 equiv.), (1) MeLi (1.1 equiv.), Et_2O, Ar, −78–0 °C, 1.5 h; (2) I_2 (1.0 equiv.), MeOH, Ar, −78 °C- r.t., 3 h; (3) NaOH (3M, aq), 0.5 h. **L**: (*R*)-**17ia** (1.0 equiv.), methyl 4-bromobenzoate (3.0 equiv.), $Pd(PPh_3)_4$ (10 mol%), Cs_2CO_3 (3.0 equiv.), THF (0.1 M), 80 °C, 50 h. **M**: (*R*)-**18ia**, (*R*)-**19ia**, or (*S*)-**20ia** (1.0 equiv.), Et_2O (5.0 equiv.), CF_3COOH (5.0 equiv.), CH_2I_2 (5.0 equiv.), DCM (0.1 M), 0 °C-r.t., 3 h.

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Table 3 | Control experiments



entry	CuTc / (*R*)-L16	[Pd(allyl)Cl]$_2$ / (*R*)-L16	Recovery of 1a (%)	Yield/ee[a] of 3aa (%)	Yield of 4aa (%) / (*Z*:*E*)	Yield of 5aa/ 5aa' (%)
1	+	+	ND	78/96	4 / (50:50)	6/17
2	–	+	92	ND/-	ND/-	ND/ND
3	+	–	39	22/0	4/(50:50)	14/23
4[b]	+	[Pd(allyl)Cl]$_2$ (0.75 equiv)/(*R*)-**L16** (1.5 equiv)	20	34/95	trace/-	trace/trace
5[c]	+	Pd(PPh$_3$)$_4$ (5 mol%)	51	3/-	26/(29:71)	ND/ND
6[c]	+	[Pd(allyl)Cl]$_2$ (2.5 mol%)/DPPF (6 mol%)	37	21/81	trace/-	11/10
7[c]	CuTc (5 mol%) / PPh$_3$ (12 mol%)	+	54	ND/-	15/(27:73)	ND/ND
8[c]	CuTc (5 mol%) / DPPF (6 mol%)	+	44	37/75	2/(50:50)	2/3

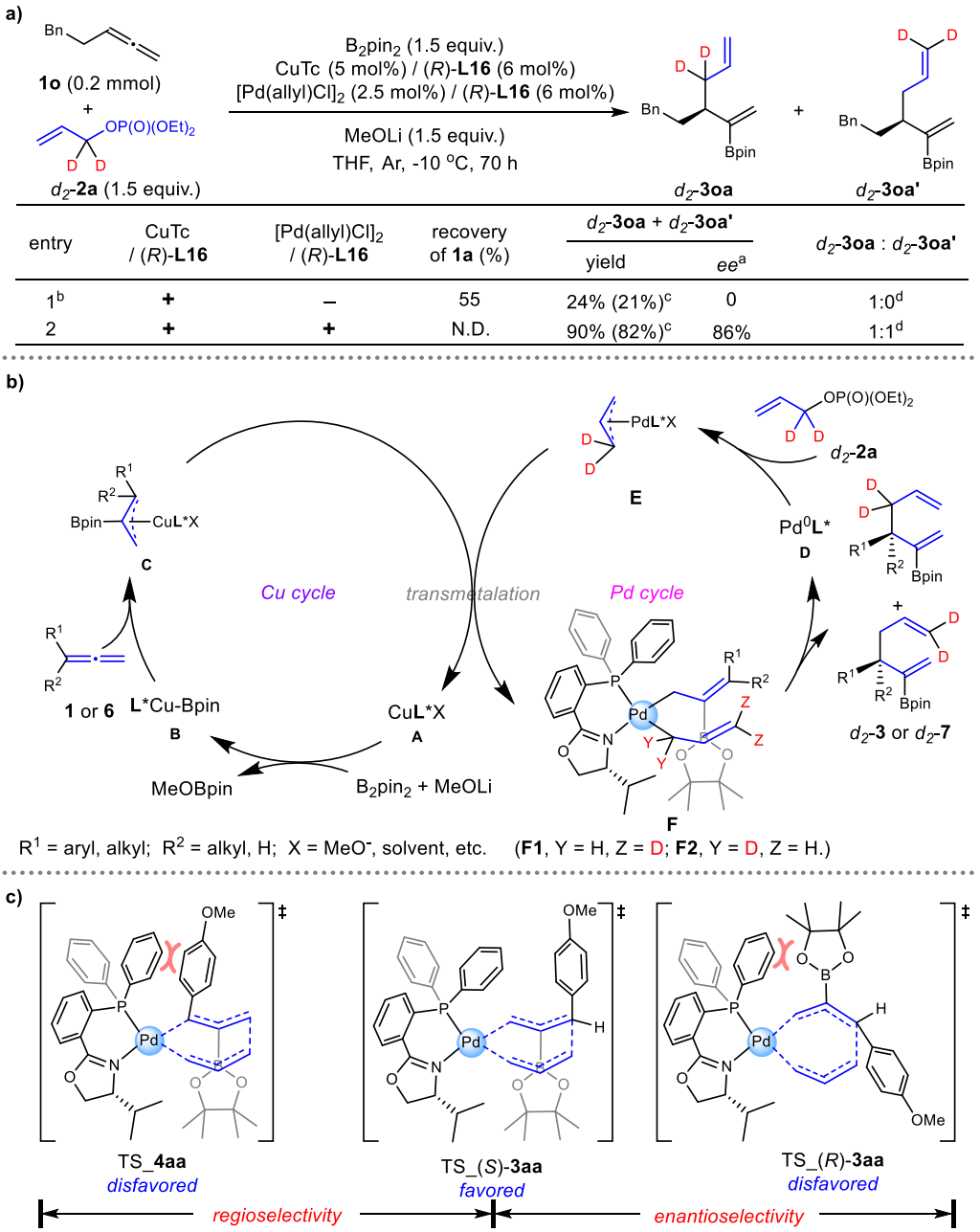
Reaction conditions: **1a** (0.2 mmol, 1.0 equiv.), **2a** (1.5 equiv.), B_2pin_2 (1.5 equiv.), MeOLi (1.5 equiv.) [Cu] (5 mol%)/(*R*)-**L16** (6 mol%), [Pd(allyl)Cl]$_2$ (2.5 mol%)/(*R*)-**L16** (6 mol%), THF (2.0 mL, 0.1 M), N_2, –10 °C, 39 h. Yields and the recoveries were determined by ^{1}H NMR analysis using CH_3NO_2 as the internal standard.
ND not detected.
[a]Determined by HPLC after oxidation of **3aa** using $NaBO_3$•$4H_2O$.
[b]Without **2a**.
[c]36 h.

cross metathesis reaction with 4-fluoro-3-phenoxystryene, chiral 1,5-diene (*R*,*E*)-**16ia** was obtained[51]. Subsequently, (*R*)-**17ia** was produced with a 60% of isolated yield and 96% ee after selective hydrogenation using Pd/C and $(MeO)_2MeSiH$ in EtOH. The C-B bond in (*R*)-**17ia** was transformed to the C-H bond upon its treatment with AgF in the mixture of THF/MeOH/H_2O, affording (*R*)-**18ia**[53]. The synthesis of (-)-protrifenbute (*R*)-**21ia** has been realized upon cyclopropanation[20]. Due to the versatile nature of the C-B bond in (*R*)-**17ia**, the derivatives of protrifenbute, (*R*)-**22ia** and (*S*)-**23ia**, were prepared with decent yields and excellent ee after coupling[49,54] and cyclopropanation.

To elucidate the mechanism, several control experiments have been performed (Table 3): no products were detected, and the starting material allene **1a** was recovered in 92% yield in the absence of CuTc/(*R*)-**L16** (Table 3, compare entry 2 with entry 1). When only chiral copper catalyst was used, the reaction yielded racemic product **3aa** in 22% yield, with **1a** being recovered with 39% (Table 3, entry 3). In the absence of allyl phosphonate **2a**, the reaction with [Pd(allyl)Cl]$_2$ (0.75 equiv.)/(*R*)-**L16** (1.5 equiv.) also afforded the target compound **3aa** with 34% yield and 95% ee, indicating that the in situ formed allyl palladium species must have provided the allyl group (Table 3, entry 4). We also carried out the reactions with the different ligands for copper and palladium. When CuTc/(*R*)-^{i}Pr-PHOX in combination with Pd(PPh$_3$)$_4$ were employed as the bimetallic catalytic system, the target product (*S*)-**3aa** was obtained in only 3% NMR yield (Table 3, entry 5). When the combination was modified to CuTc/(*R*)-^{i}Pr-PHOX and [Pd(allyl)Cl]$_2$/DPPF, (*S*)-**3aa** was obtained in 21% NMR yield with 81% ee (Table 3, entry 6). When PPh$_3$ was employed as the ligand for CuTc, (*S*)-**3aa** was not detected (Table 3, entry 7). When DPPF was applied for CuTc, (*S*)-**3aa** was obtained in 37% NMR yield with 75% ee (Table 3, entry 8). Thus, the optimal chiral ligand (*R*)-^{i}Pr-PHOX was necessary for both copper and palladium catalysts. In order to further demonstrate the nature of allyl metallic intermediate in this synergistic Cu/Pd catalytic cycles, isotopic labeling experiments were designed and conducted (Fig. 5a): with CuTc/(*R*)-**L16** as the catalyst precursors, only racemic d_2-**3oa** was produced with 21% isolated yield, which is consistent with the above results (Fig. 5a, entry 1 and Table 3, entry 3), and the absence of d_2-**3oa'** as judged by ^{1}H NMR analysis of isolated products indicated that copper catalytic cycle must have proceeded through S_N2-type process with d_2-**2a** to produce the racemic product. However, when synergistic Cu/Pd catalysts were used, the desired products d_2-**3oa** and d_2-**3oa'** were obtained in 1:1 ratio and a combined isolated yield of 82% and enantioselectivity of 86% ee, which provides clear evidence for the involvement of η^3-allyl palladium species in the catalytic cycles and reveals the necessity of the Pd catalysis for this asymmetric allylboration reaction.

Based on control experiments, catalytic cycles of synergistic Cu/Pd regio- and enantioselective catalysis have been proposed (Fig. 5b): Firstly, chiral Bpin-Cu[(*R*)-**L16**] species **B** was generated via the transmetalation between B_2pin_2 and (*R*)-**L16** coordinated copper complex **A**. The η^3-allylic copper species **C** was generated via the borylcupration of allenes, which may react with the in situ generated chiral π-allylpalladium species **E** via the oxidative addition of allyl diethyl phosphate d_2-**2a** with Pd^0[(*R*)-**L16**] complex to afford bis(allyl) palladium complex **F**. Due to the nature of the Pd^{2+} and the 18 e electron rule, it is believed that most likely the η^3-allylic unit was transferred from the Cu atom to the Pd atom to afford η^1-allylic species **F1** and **F2**[55–57]. Finally, target products d_2-**3** or d_2-**7** with deuterium atom exclusively at either end of the three carbon atoms from the deuterated allyl phosphonate d_2-**2a** were obtained upon reductive elimination of these two palladium intermediates **F1** and **F2** due to steric effect for pushing more sterically hindered terminal away from this *non*-C_2-symmetric P, N-ligand (*R*)-**L16**, resulting in an excellent regio- and enantioselectivity, which may be rationalized by the bigger steric hindrance in TS_(*R*)-**3aa** and TS_**4aa**, thus, favoring the formation of (*S*)-**3aa** by the reductive elimination via TS_(*S*)-**3aa** (Fig. 5c).

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entry	CuTc / (*R*)-**L16**	[Pd(allyl)Cl]$_2$ / (*R*)-**L16**	recovery of **1a** (%)	d_2-**3oa** + d_2-**3oa'** yield	d_2-**3oa** + d_2-**3oa'** *ee*[a]	d_2-**3oa** : d_2-**3oa'**
1[b]	+	–	55	24% (21%)[c]	0	1:0[d]
2	+	+	N.D.	90% (82%)[c]	86%	1:1[d]

Fig. 5 | Isotopic labeling experiments and proposed reaction mechanism. **a** Isotopic labeling experiments. Yields and the recoveries were determined by ^{1}H NMR analysis using CH_3NO_2 as the internal standard. [a] determined by HPLC analysis after oxidation of d_2-**3oa**/**3oa'** using $NaBO_3$•$4H_2O$. [b] The reaction was conducted on a 0.5 mmol scale at 0.25 M. [c] isolated yield. [d] determined by ^{1}H NMR analysis of isolated products. ND not detected. **b** Proposed reaction mechanism. **c** Proposed rationale for the regio- and enantioselectivity.

In summary, synergistic Cu/Pd catalyzed three-component branched regioselective asymmetric allylboration of allenes with B_2pin_2 and allylic phosphates has been developed, affording a series of chiral 1,5-dienes containing a sp^2 C-B bond with moderate to good yields. Both mono- or 1,1-disubstituted terminal allenes substituted with functional groups are ideal partners. Due to the versatile nature of the products, a set of chiral molecules may be formed efficiently. Synthesis of (-)-protrifenbute and the preparation of its derivatives have been realized upon using this protocol as the key step. Control experiments reveal that the synergistic Cu/Pd catalysis with *non*-C_2-symmetric phosphine-oxazoline chiral ligand **L16** is responsible for the regulation of the regio- and enantioselectivity in the current bifunctionalization reaction of allenes.

Methods

General procedure for the synthesis of allylboration products 3 or 7

Preparation of the Cu pre-catalyst in a nitrogen-filled glovebox: to an oven-dried 4 mL vial were added a stirring bar, CuTc (9.5 mg, 0.05 mmol), (*R*)-**L16** (22.4 mg, 0.06 mmol), and THF (1.0 mL). The resulting mixture was stirred at room temperature for 10 min to afford the Cu pre-catalyst, which was transferred via a 1.0 mL syringe.

Preparation of the Pd pre-catalyst in a nitrogen-filled glovebox: to an oven-dried 4 mL vial were added a stirring bar, [Pd(allyl)Cl]$_2$ (9.2 mg, 0.025 mmol), (*R*)-**L16** (22.4 mg, 0.06 mmol), and THF (1.0 mL). The resulting mixture was stirred at room temperature for 10 min to afford the Pd pre-catalyst, which was transferred via a 1.0 mL syringe.

In a nitrogen-filled glovebox, to an oven-dried 25 mL tube with screw cap were added a stirring bar, B_2Pin_2, MeOLi, allene **1** or **6**, allylic phosphates **2**, and THF (2.0 mL). The tube with a screw cap and Cu/Pd pre-catalysts in syringes were then taken out of the glovebox together. The sealed tube was connected to the Schlenk line and degassed/refilled with argon three times. Then, the sealed tube was immersed in the EtOH bath controlled at −10 °C. Under argon atmosphere, the screw cap was removed, and the Pd pre-catalyst and Cu pre-catalyst were added by syringes sequentially and the tube was screwed tightly. The resulting mixture was vigorously stirred at −10 °C as monitored by TLC and filtered over a plug of silica gel eluted with ethyl acetate (50 mL). The crude product was purified by column chromatography on silica gel to afford the product **3** or **7**.

Data availability

Experimental procedures and characterization of the compounds are available in the Supplementary Information. Crystallographic data have been deposited at the Cambridge Crystallographic Data Center, under deposition numbers CCDC 2373589 ((*R*)-**7ea**) and CCDC 2373590 ((*R,E*)-**13ca**). Copies of the data can be obtained free of charge via https://www.ccdc.cam.ac.uk/structures/. All other data are available from the corresponding authors upon request.

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Author contributions

S.M. and H.Q. directed the research. S.M., H.Q., and Z.X. designed the experiments. Z.X. performed the experiments. Z.X., H.Q., and S.M. wrote the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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